



Machine Learning (CSE543/ECE543) Midsem (20 Marks) (Duration 60 min)



• Beginner

4 + 4 points

1. An artificially intelligent car decreases its speed based on its distance from the car in front of it. Which algorithm the car might be using? Explain your reasoning.
(a) SVMs (b) Decision Tree (c) Linear Regression (d) Logistic Regression
2. Explain the difference between Parametric and Non-parametric learning methods. Are Decision Tree parametric? Explain.

• Intermediate

4 points

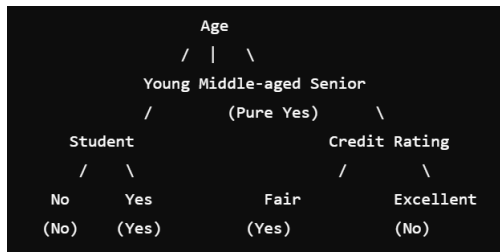
1. Derive weight (slope and intercept) update equation for linear regression using mean absolute error (MAE) loss function.

$$MAE = \frac{1}{n} \sum_i |y_i - \hat{y}_i|$$

• Advanced

8 points

1. Build a Decision Tree Using Gini Gain as the splitting criterion to predict whether a person will purchase a new car.



ID	Age	Income	Student	Credit Rating	Buy Car
1	Young	High	No	Fair	No
2	Young	High	No	Excellent	No
3	Middle-aged	High	No	Fair	Yes
4	Senior	Medium	No	Fair	Yes
5	Senior	Low	Yes	Fair	Yes
6	Senior	Low	Yes	Excellent	No
7	Middle-aged	Low	Yes	Excellent	Yes
8	Young	Medium	No	Fair	No
9	Young	Low	Yes	Fair	Yes
10	Senior	Medium	Yes	Fair	Yes
11	Young	Medium	Yes	Excellent	Yes
12	Middle-aged	Medium	No	Excellent	Yes
13	Middle-aged	High	Yes	Fair	Yes
14	Senior	Medium	No	Excellent	No